# New Wheels Project

## Introduction to SQL

## Problem Statement

### Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

### Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

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# Business Questions

## Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

**Solution Query:**

```
SELECT c.state,
    COUNT(DISTINCT o.customer_id) AS customers_count
FROM order_t o
JOIN customer_t c ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY customers_count DESC, c.state;
```

**Output:**



Query:

```
SELECT c.state,
    COUNT(DISTINCT o.customer_id) AS customers_count
FROM order_t o
JOIN customer_t c ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY customers_count DESC, c.state
```

Output:

Showing first 10 rows out of 49 rows

| state | customers_count |
|---|---|
| California | 97 |
| Texas | 97 |
| Florida | 86 |
| New York | 69 |
| District of Columbia | 35 |
| Colorado | 33 |
| Ohio | 33 |
| Alabama | 29 |
| Washington | 28 |
| Arizona | 26 |

**Observations and Insights:**

- The customer base is concentrated in a few high-volume states, with California and Texas tied at the top (97 each), followed by Florida (86).

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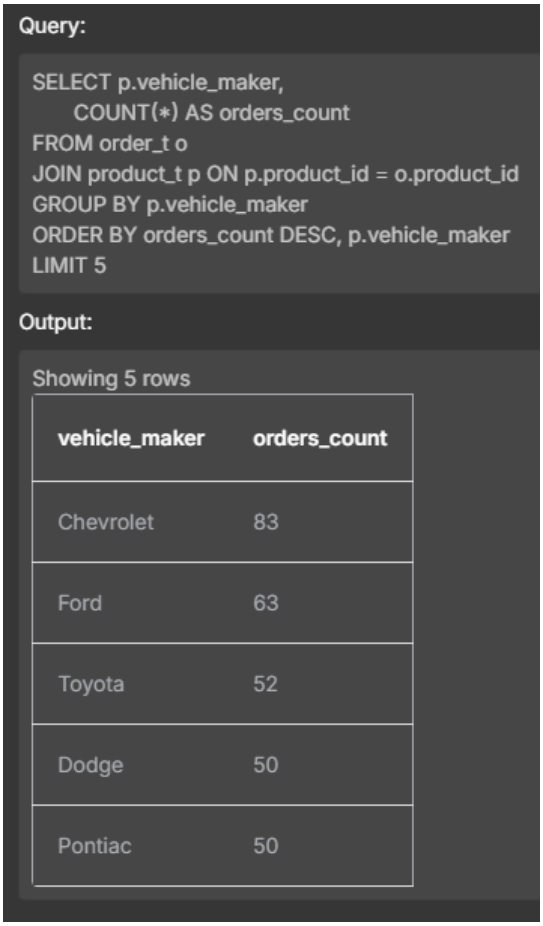
- This suggests the company should prioritize demand planning, targeted campaigns, and service coverage in these top states.
- Mid-tier states such as New York and District of Columbia still contribute meaningfully, indicating room for regional growth beyond the top 3.

## Question 2: Which are the top 5 vehicle makers preferred by the customers?

**Solution Query:**

```
SELECT p.vehicle_maker,
    COUNT(*) AS orders_count
FROM order_t o
JOIN product_t p ON p.product_id = o.product_id
GROUP BY p.vehicle_maker
ORDER BY orders_count DESC, p.vehicle_maker
LIMIT 5;
```

**Output:**




**Observations and Insights:**

- Chevrolet (83) and Ford (63) are the most preferred makers, with a noticeable lead over the rest.

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- The remaining top brands (Toyota, Dodge, Pontiac) are relatively close, indicating competition in the second tier.
- Inventory and promotions can be weighted toward top-performing makers while preserving a diversified catalog for customer choice.

## Question 3: Which is the most preferred vehicle maker in each state?

**Solution Query:**

```
SELECT state, vehicle_maker, customer_count
FROM (
   SELECT
      c.state,
      p.vehicle_maker,
      COUNT(DISTINCT o.customer_id) AS customer_count,
      ROW_NUMBER() OVER (
         PARTITION BY c.state
         ORDER BY COUNT(DISTINCT o.customer_id) DESC, p.vehicle_maker
      ) AS rn
   FROM order_t o
   JOIN customer_t c ON c.customer_id = o.customer_id
   JOIN product_t p ON p.product_id = o.product_id
   GROUP BY c.state, p.vehicle_maker
) t
WHERE rn = 1
ORDER BY state;
```

**Output:**

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Query:

```
SELECT state, vehicle_maker, customer_count
FROM (
   SELECT
      c.state,
      p.vehicle_maker,
      COUNT(DISTINCT o.customer_id) AS customer_count,
      ROW_NUMBER() OVER (
         PARTITION BY c.state
         ORDER BY COUNT(DISTINCT o.customer_id) DESC, p.vehicle_maker
      ) AS rn
   FROM order_t o
   JOIN customer_t c ON c.customer_id = o.customer_id
   JOIN product_t p ON p.product_id = o.product_id
   GROUP BY c.state, p.vehicle_maker
) t
WHERE rn = 1
ORDER BY state
```

Output:

Showing first 10 rows out of 49 rows

| state | vehicle_maker | customer_count |
|---|---|---|
| Alabama | Dodge | 5 |
| Alaska | Chevrolet | 2 |
| Arizona | Cadillac | 3 |
| Arkansas | Chevrolet | 1 |
| California | Audi | 6 |
| Colorado | Chevrolet | 5 |
| Connecticut | Chevrolet | 2 |
| Delaware | Mitsubishi | 2 |
| District of Columbia | Chevrolet | 4 |

**Observations and Insights:**

- Preferred makers vary by state, which confirms that demand is region-specific rather than nationally uniform.
- Some brands (for example, Chevrolet in multiple states) appear repeatedly, indicating broad acceptance across geographies.
- The company should use state-level preference mapping to optimize assortment, local campaigns, and fulfillment decisions.

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**Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?**

**Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5**

**Solution Query:**

```
SELECT
   q.quarter_number,
   ROUND(q.avg_rating_quarter, 2) AS avg_rating_quarter,
   ROUND(o.overall_avg_rating, 2) AS overall_avg_rating
FROM (
   SELECT
      quarter_number,
      AVG(
         CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad' THEN 2
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
         END
      ) AS avg_rating_quarter
   FROM order_t
   GROUP BY quarter_number
) q
CROSS JOIN (
   SELECT
      AVG(
         CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad' THEN 2
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
         END
      ) AS overall_avg_rating
   FROM order_t
) o
ORDER BY q.quarter_number;
```

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**Output:**

Query:

```
SELECT
  q.quarter_number,
  ROUND(q.avg_rating_quarter, 2) AS avg_rating_quarter,
  ROUND(o.overall_avg_rating, 2) AS overall_avg_rating
FROM (
  SELECT
    quarter_number,
    AVG(
      CASE customer_feedback
        WHEN 'Very Bad' THEN 1
        WHEN 'Bad' THEN 2
        WHEN 'Okay' THEN 3
        WHEN 'Good' THEN 4
        WHEN 'Very Good' THEN 5
      END
    ) AS avg_rating_quarter
  FROM order_t
  GROUP BY quarter_number) q
CROSS JOIN (
  SELECT
    AVG(
      CASE customer_feedback
        WHEN 'Very Bad' THEN 1
        WHEN 'Bad' THEN 2
        WHEN 'Okay' THEN 3
        WHEN 'Good' THEN 4
        WHEN 'Very Good' THEN 5
      END
    ) AS overall_avg_rating
  FROM order_t) o
ORDER BY q.quarter_number
```

Output:

Showing 4 rows

| quarter_number | avg_rating_quarter | overall_avg_rating |
|---|---|---|
| 1 | 3.55 | 3.14 |
| 2 | 3.35 | 3.14 |
| 3 | 2.96 | 3.14 |
| 4 | 2.4 | 3.14 |

**Observations and Insights:**

- The overall average customer rating is 3.14/5, indicating moderate satisfaction with meaningful scope for service improvement.
- Quarterly average ratings declined steadily from 3.55 (Q1) to 2.40 (Q4), showing a clear deterioration in customer experience over time.
- The gap between Q1 and Q4 suggests growing dissatisfaction, which may impact repeat purchases, referrals, and overall brand trust if not addressed quickly.

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## Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

**Solution Query:**

```
SELECT quarter_number,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) /
COUNT(*), 2) AS pct_very_bad,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS pct_bad,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS pct_okay,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS pct_good,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) /
COUNT(*), 2) AS pct_very_good
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

**Output:**

Query:

```
SELECT quarter_number,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) / COUNT(*), 2) AS pct_very_bad,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) / COUNT(*), 2) AS pct_bad,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) / COUNT(*), 2) AS pct_okay,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) / COUNT(*), 2) AS pct_good,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) / COUNT(*), 2) AS pct_very_good
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

| quarter_number | pct_very_bad | pct_bad | pct_okay | pct_good | pct_very_good |
|---|---|---|---|---|---|
| 1 | 10.97 | 11.29 | 19.03 | 28.71 | 30 |
| 2 | 14.89 | 14.12 | 20.23 | 22.14 | 28.63 |
| 3 | 17.9 | 22.71 | 21.83 | 20.96 | 16.59 |
| 4 | 30.65 | 29.15 | 20.1 | 10.05 | 10.05 |

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**Observations and Insights:**

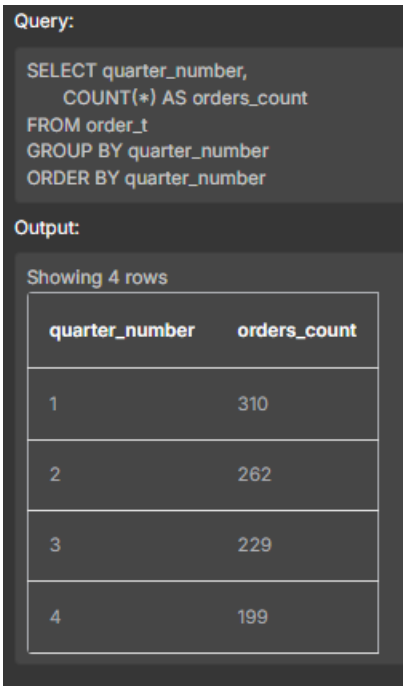
- In Q1, positive feedback is strongest (Good + Very Good = 58.71%) while negative feedback (Very Bad + Bad = 22.26%) is comparatively low.
- By Q4, this trend reverses sharply: negative feedback rises to 59.80%, while positive feedback drops to 20.10%.
- This quarter-wise shift confirms that customers are becoming increasingly dissatisfied over time, likely reflecting worsening service/fulfillment experience.

## Question 6: What is the trend of the number of orders by quarter?

**Solution Query:**

```
SELECT quarter_number,
      COUNT(*) AS orders_count
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

**Output:**




**Observations and Insights:**

- Quarterly orders declined consistently from 310 (Q1) to 199 (Q4), showing a clear downward demand trend.
- The drop is continuous across all quarters (310 -> 262 -> 229 -> 199), with no quarter showing recovery.

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- This sustained decline suggests weakening customer traction over time and highlights the need for immediate retention and service-quality actions.

## Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

**Solution Query:**

```
SELECT
    curr.quarter_number,
    ROUND(curr.net_revenue, 2) AS net_revenue,
    ROUND(
        100.0 * (curr.net_revenue - prev.net_revenue) / prev.net_revenue,
        2
    ) AS qoq_pct_change
FROM (
    SELECT
        quarter_number,
        SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
    FROM order_t
    GROUP BY quarter_number
) curr
LEFT JOIN (
    SELECT
        quarter_number,
        SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
    FROM order_t
    GROUP BY quarter_number
) prev
    ON curr.quarter_number = prev.quarter_number + 1
ORDER BY curr.quarter_number;
```

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**Output:**

Query:

```
SELECT
    curr.quarter_number,
    ROUND(curr.net_revenue, 2) AS net_revenue,
    ROUND(
        100.0 * (curr.net_revenue - prev.net_revenue) / prev.net_revenue,
        2
    ) AS qoq_pct_change
FROM (
    SELECT
        quarter_number,
        SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
    FROM order_t
    GROUP BY quarter_number
) curr
LEFT JOIN (
    SELECT
        quarter_number,
        SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
    FROM order_t
    GROUP BY quarter_number
) prev
    ON curr.quarter_number = prev.quarter_number + 1
ORDER BY curr.quarter_number
```

Output:

Showing 4 rows

| quarter_number | net_revenue | qoq_pct_change |
|---|---|---|
| 1 | 18032549.9 | |
| 2 | 13122995.76 | -27.23 |
| 3 | 8882298.84 | -32.32 |
| 4 | 8573149.28 | -3.48 |

**Observations and Insights:**

- Net revenue declines every quarter: 18,032,549.90 (Q1) -> 13,122,995.76 (Q2) -> 8,882,298.84 (Q3) -> 8,573,149.28 (Q4).
- QoQ change is negative after Q1 (baseline): -27.23% (Q2), -32.32% (Q3), and -3.48% (Q4).
- The steepest drop occurs in Q3, while Q4 shows a smaller decline, indicating the fall is slowing but revenue remains significantly below Q1.

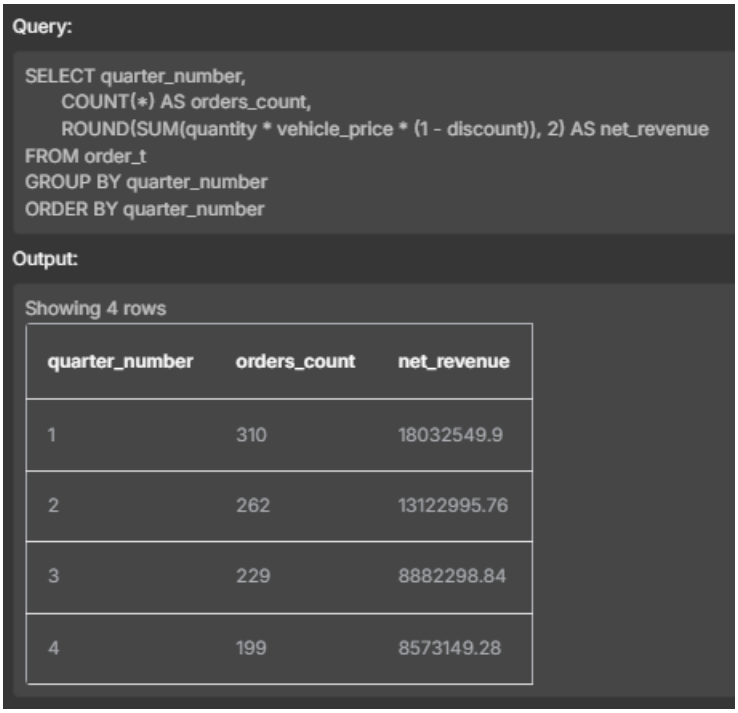
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## Question 8: What is the trend of net revenue and orders by quarters?

**Solution Query:**

```
SELECT quarter_number,
    COUNT(*) AS orders_count,
    ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

**Output:**



Query:

```
SELECT quarter_number,
    COUNT(*) AS orders_count,
    ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

| quarter_number | orders_count | net_revenue |
|---|---|---|
| 1 | 310 | 18032549.9 |
| 2 | 262 | 13122995.76 |
| 3 | 229 | 8882298.84 |
| 4 | 199 | 8573149.28 |

**Observations and Insights:**

- Both KPIs decline quarter-over-quarter: orders fall from 310 (Q1) to 199 (Q4), while net revenue drops from 18,032,549.90 to 8,573,149.28.
- The most pronounced revenue reduction is between Q2 and Q3 (13,122,995.76 to 8,882,298.84), indicating a major mid-year slowdown.
- Q4 records the lowest orders and revenue, confirming sustained weakness in both demand and overall business performance.

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## Question 9: What is the average discount offered for different types of credit cards?

**Solution Query:**

```
SELECT c.credit_card_type,
    ROUND(AVG(o.discount) * 100, 2) AS avg_discount_pct
FROM order_t o
JOIN customer_t c ON c.customer_id = o.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount_pct DESC, c.credit_card_type;
```

**Output:**

Query:

```
SELECT c.credit_card_type,
    ROUND(AVG(o.discount) * 100, 2) AS avg_discount_pct
FROM order_t o
JOIN customer_t c ON c.customer_id = o.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount_pct DESC, c.credit_card_type
```

Output:

Showing first 10 rows out of 16 rows

| credit_card_type | avg_discount_pct |
|---|---|
| laser | 64.38 |
| mastercard | 62.95 |
| maestro | 62.42 |
| visa-electron | 62.35 |
| china-unionpay | 62.22 |
| instapayment | 62.06 |
| americanexpress | 61.63 |
| diners-club-us-ca | 61.46 |
| diners-club-carte-blan... | 61.45 |
| switch | 61.02 |

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**Observations and Insights:**

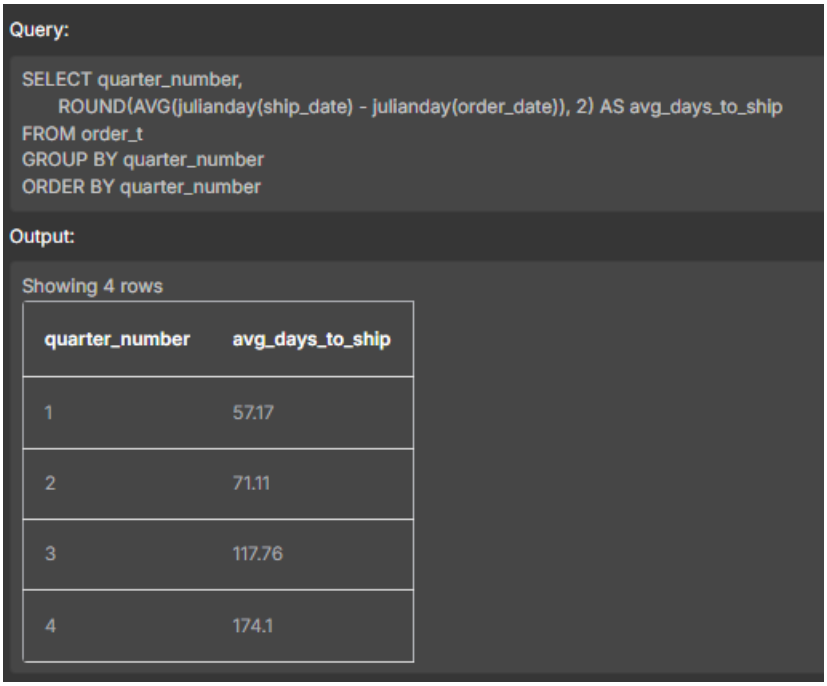
- Discount levels are high across card types; among the results displayed, laser has the highest average discount at 64.38%.
- Other leading card types (mastercard 62.95%, maestro 62.42%, visa-electron 62.35%, china-unionpay 62.22%) are close, indicating a fairly narrow discount spread.
- Since the output shows 10 of 16 card types, the overall pattern suggests discounting is broadly consistent rather than concentrated in just one payment segment.

## Question 10: What is the average time taken to ship the placed orders for each quarter?

**Solution Query:**

```
SELECT quarter_number,
     ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS avg_days_to_ship
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

**Output:**




**Observations and Insights:**

- Average shipping time increases quarter by quarter from 57.17 days (Q1) to 174.10 days (Q4).
- The upward trend (57.17 -> 71.11 -> 117.76 -> 174.10) indicates progressively slower deliveries across the year.

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- This sharp rise in fulfillment time is likely a key driver of lower customer satisfaction and declining business performance in later quarters.

## Business Metrics Overview

| Total Revenue | Total Orders | Total Customers | Average Rating |
|---|---|---|---|
| 48610993.78 | 1000 | 994 | 3.07 |
| **Last Quarter Revenue** | **Last quarter Orders** | **Average Days to Ship** | **% Good Feedback** |
| 8573149.28 | 199 | 174.1 | 20.1% |

## Business Recommendations

- Fix fulfillment performance first: Launch a 90-day logistics improvement plan to reduce average shipping days, with quarterly SLA targets and route-level monitoring.
- Recover customer satisfaction proactively: Introduce post-delivery follow-ups, faster complaint resolution, and service recovery offers for low-rating customers to reverse the Q4 feedback decline.
- Protect revenue while optimizing discounts: Audit card-type discount effectiveness and shift from broad high discounting to targeted promotions that improve conversion without eroding margins.
- Focus growth where demand is strongest: Prioritize inventory, marketing spend, and partner coverage in top states and top vehicle makers, while running localized campaigns in underperforming regions.
- Track business health through a quarterly control dashboard: Monitor orders, net revenue, rating score, negative feedback %, and average shipping days together to trigger early corrective action.
- Total Revenue: Set a quarterly revenue recovery plan with maker/state-level targets and prioritize high-margin products to improve overall top-line performance.
- Total Orders: Launch conversion and retention campaigns (remarketing, exchange offers, referral incentives) to reverse the declining order trajectory.
- Total Customers: Build repeat-customer programs and segmented communication journeys to improve customer lifetime value and reduce churn risk.
- Average Rating: Establish a closed-loop feedback system for all low ratings (1-2), with 48-hour resolution SLAs and root-cause tracking.
- Last Quarter Revenue (Q4): Run a Q4 recovery playbook with targeted seasonal bundles and tighter discount controls to lift quarter-end monetization.
- Last Quarter Orders (Q4): Strengthen demand generation in top-performing states and re-engage inactive customers to improve immediate quarter volume.
- Average Days to Ship: Treat fulfillment delay as a critical KPI; optimize shipper mix, warehouse allocation, and dispatch cadence to reduce delivery time.

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- % Good Feedback: Increase good-feedback share through delivery reliability, proactive status updates, and post-delivery service assurance.

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